Asymptotic constants and finite exclusion tests for the number of alkane isomers:

computational notes on the sequence A000602

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Abstract

Let $a(n)$ denote the number of structural isomers of the alkane C_nH_{2n+2}, i.e. the number of unlabeled, unrooted trees on n vertices with maximum degree 4 (OEIS A000602). No exact closed form for $a(n)$ is known. We study several standard mechanisms that often lead to exact formulas for counting sequences, and give finite exact-arithmetic tests around them. Specifically, we prove by rigorous modular rank certificates, validated against 621 exactly computed terms (cross-checked against the independently computed OEIS *b*-file), that: (i) $a(n)$ satisfies *no* polynomial-coefficient recurrence, homogeneous or inhomogeneous, in a staircase region extending to order 2/degree 150, order 10/degree 46, and order 304 with constant coefficients — consequently no fixed-depth proper hypergeometric multisum closed form (Wilf–Zeilberger class) with canonical annihilator in that region can equal $a(n)$; (ii) the generating function satisfies no algebraic equation of y-degree ≤ 28 within an analogous degree staircase, and no first-order algebraic differential equation of degree ≤ 7 in each of x, y, y'; (iii) $a(n)$ mod 2 has 2-kernel of size ≥ 31 (maximal growth through depth 4) and $a(n)$ mod m is not eventually periodic with period ≤ 200 for $2 \leq m \leq 7$, giving finite evidence against small digit-automatic descriptions. We then prove that the *labeled* analogue of the problem admits an exact two-index finite-sum closed form, emphasizing that the unlabeled quotient is the main source of difficulty: Burnside's lemma expresses $a(n)$ as a sum over the partitions of n, an index set of unbounded dimension. Finally we show structurally why the best-known mechanism for collapsing such sums to finite exact formulas — Rademacher's modular circle method — has no evident analogue here: the generating function is governed by a mixed $(2,3)$-Mahler-type functional system whose singularities cascade toward a (conjecturally natural) boundary on the unit circle without any known comparable transformation theory. Together these results give explicit finite constraints and a structural picture that any future closed form would have to address.

Contents

1 Introduction

1.1 The problem

For $n \geq 1$ let $a(n)$ denote the number of isomorphism classes of trees on n unlabeled vertices in which every vertex has degree at most 4. Chemically, $a(n)$ is the number of constitutional (structural) isomers of the alkane C_nH_{2n+2}, stereoisomers not distinguished. The sequence begins (with the convention $a(0) = 1$)

$$1,\ 1,\ 1,\ 1,\ 2,\ 3,\ 5,\ 9,\ 18,\ 35,\ 75,\ 159,\ 355,\ 802,\ 1858,\ 4347,\ \ldots$$

and is sequence A000602 of the OEIS [11]. Its enumeration, initiated by Cayley in 1875 [3] and completed as an *algorithm* by Pólya in 1937 [14], was the founding application of Pólya enumeration theory. Yet after a century and a half the following question is open.

Question 1.1 (Problem (A))**.** Does there exist a function $F(n)$, built from finitely many elementary operations $(+, -, \times, \div)$, exponentials and logarithms, standard special functions (e.g. Γ, ζ, hypergeometric ${}_pF_q$), and *finite* (fixed-depth) sums and products — directly evaluable at any n without knowledge of $a(1), \ldots, a(n-1)$, and not an implicit functional equation or a coefficient-extraction from a generating function — such that $F(n) = a(n)$ for all $n \geq 1$?

This note does not answer Question 1.1. Its purpose is to demonstrate, with certificates that are rigorous modulo only the correctness of widely validated exact-arithmetic computations, that several standard routes to such a formula fail in substantial finite ranges, to quantify those ranges, and to record the structural source of difficulty suggested by the generating functions. All computations are reproducible from the scripts described in Appendix A.

1.2 Why the question is sharp

Two remarks calibrate what "closed form" can possibly mean here.

Proposition 1.2 (Additive approximation is equivalent to exactness)**.** *Suppose F is directly evaluable and $|F(n) - a(n)| < \frac{1}{2}$ for all $n \geq n_0$. Then $\lfloor F(n) + \frac{1}{2} \rfloor$ is an exact closed form for $a(n)$ for $n \geq n_0$. Hence "closed form with additive error $O(n^{-k})$" is not a relaxation of Question 1.1 but a reformulation of it.*

Proof. Immediate from $a(n) \in \mathbb{Z}$. □

Remark 1.3. Question 1.1 is *not* vacuous for sequences of this general type: Rademacher's convergent series for the partition function $p(n)$ [15], truncated at $O(\sqrt{n})$ terms with a certified error below $\frac{1}{2}$, is exactly such a directly evaluable formula, even though the generating function $\prod(1 - x^k)^{-1}$ has the unit circle as a natural boundary. The mechanism enabling it — modularity of the Dedekind eta function — will reappear in §8 as the kind of structure not known for $a(n)$.

By contrast, the *multiplicative* (relative-error) version of the question is completely solved by classical singularity analysis: Pólya [14] proved $a(n) \sim C\,\alpha^n n^{-5/2}$, and the expansion extends to every order (see §8),

$$a(n) \;=\; C\,\alpha^n\, n^{-5/2}\Big(1 + \frac{c_1}{n} + \frac{c_2}{n^2} + \cdots + \frac{c_m}{n^m} + O(n^{-m-1})\Big), \tag{1}$$

with $\alpha = \rho^{-1} = 2.8154600331761507465266167 78\ldots$, $C = 0.6563186958418347 50\ldots$, $c_1 = 0.03006869504\ldots$, $c_2 = 4.7882749807\ldots$, $c_3 \approx 22.18$ (constants computed in this project; α and C classical [17, 6, 7]). The gap between (1) and Question 1.1 is exponential: by Proposition 1.2 an exact formula needs relative accuracy $\alpha^{-n} n^{5/2}$, while any fixed truncation of (1) achieves only polynomial relative accuracy.

1.3 Summary of results

Write $\mathcal{R}(r, d)$ for the statement "there exist polynomials $p_0, \ldots, p_r, q \in \mathbb{C}[n]$ of degree $\leq d$, with $(p_0, \ldots, p_r) \neq 0$, such that $\sum_{j=0}^{r} p_j(n)\, a(n + j) = q(n)$ for all $n \geq 1$".

Theorem 1.4 (Non-holonomicity perimeter; §4)**.** $\mathcal{R}(r, d)$ *is* false *for every pair* (r, d) *in the staircase region of Table 1, which contains in particular* $(2, 150)$, $(3, 120)$, $(4, 100)$, $(10, 46)$, $(56, 8)$, $(110, 3)$, $(200, 1)$ *and* $(304, 0)$.

Theorem 1.5 (Non-algebraicity perimeter; §5)**.** *The generating function $A(x) = \sum_{n \geq 0} a(n)x^n$ satisfies no nonzero polynomial relation $P(x, A(x)) = 0$ with $\deg_y P \leq D_y$, $\deg_x P \leq D_x$ for any (D_y, D_x) in the staircase of Table 2 (up to $(28, 19)$ and $(2, 190)$), and no relation $P(x, A(x), A'(x)) = 0$ with degree ≤ 7 in each variable.*

Theorem 1.6 (Arithmetic obstructions; §6)**.** *The 2-kernel of $(a(n) \bmod 2)$ has at least 31 pairwise distinct elements (its growth through depth 4 is the maximum possible, $2^{k+1} - 1$), and the 3-kernel of $(a(n) \bmod 3)$ has at least 40. For every modulus $2 \leq m \leq 7$, $(a(n) \bmod m)$ is not eventually periodic with any period ≤ 200.*

Theorem 1.7 (The labeled problem is closed-form; §7)**.** *The number $L(n)$ of* labeled *trees on $n \geq 3$ vertices with maximum degree 4 equals the two-index finite sum*

$$L(n) = \sum_{d=0}^{\lfloor (n-2)/3 \rfloor} \sum_{c=0}^{\lfloor (n-2-3d)/2 \rfloor} \frac{n!\,(n-2)!}{a!\,b!\,c!\,d!\,2^c\,6^d}, \qquad b = n-2-2c-3d, \quad a = n-b-c-d,$$

which lies squarely in the function class of Question 1.1. Hence the degree-4 (valence) constraint is not *the obstruction; the unlabeled quotient is.*

Section 8 assembles a structural picture: the Burnside sum over partitions, the Otter cancellation that produces the exponent $-5/2$, the singularity cascade of the underlying mixed $(2,3)$-Mahler system, and the absence of an evident Rademacher mechanism. Section 9 states the resulting constraints on possible solutions of Question 1.1.

Epistemic status

Theorems 1.4–1.6 are computer-assisted. The underlying linear algebra is exact (machine-word arithmetic modulo primes $< 2^{26}$, no floating point); the logical step from a full-rank computation to non-existence is a one-line lemma (Lemma 3.1); the data feeding the matrices is verified in §2. Implementation risk is controlled by decoy sequences with known positive answers (the guessers *do* find the Catalan recurrence, the Catalan algebraic equation, and the Thue–Morse kernel) and by re-verification modulo a second prime. The propositions of §8 marked "sketch" are structural/heuristic and clearly flagged; nothing in §§4–7 depends on them.

2 The defining system and certified data

2.1 The Pólya–Otter system

Let Z_{S_3} and Z_{S_4} be the cycle index polynomials

$$Z_{S_3}(t_1,t_2,t_3) = \tfrac{1}{6}\big(t_1^3 + 3t_1t_2 + 2t_3\big), \qquad Z_{S_4}(t_1,t_2,t_3,t_4) = \tfrac{1}{24}\big(t_1^4 + 6t_1^2t_2 + 3t_2^2 + 8t_1t_3 + 6t_4\big).$$

Let $T(x) = \sum_{n\geq 0} t(n)x^n$ be the generating function of A000598 (alkyl radicals: rooted trees in which every vertex has at most 3 children), defined by

$$T(x) \;=\; 1 + x\,Z_{S_3}\big(T(x), T(x^2), T(x^3)\big) \;=\; 1 + \frac{x}{6}\Big(T(x)^3 + 3\,T(x)\,T(x^2) + 2\,T(x^3)\Big). \tag{2}$$

The unrooted count is recovered by the dissymmetry (Otter) formula [12, 8, 16]:

$$A(x) \;=\; \sum_{n\geq 0} a(n)x^n \;=\; 1 + x\,Z_{S_4}\big(T(x), T(x^2), T(x^3), T(x^4)\big) - \frac{\big(T(x)-1\big)^2 - \big(T(x^2)-1\big)}{2}. \tag{3}$$

Equations (2)–(3) are the standard formulation (they appear, in equivalent form, on the OEIS pages of A000598 and A000602); they constitute a *recursive* description, not a closed form: extracting $a(n)$ requires the n preceding coefficients of T.

2.2 Exact computation and certification

Working in exact integer arithmetic, the coefficients $t(0), \ldots, t(620)$ were computed from (2) by incremental convolution (the divisions by 6 are exact at every step, a useful integrality check that was asserted at each n), and $a(0), \ldots, a(620)$ from (3). The value $a(620)$ has 272 decimal digits.

Certification of the data.

1. $t(0..14)$ and $a(0..20)$ agree with the values printed in the OEIS entries A000598, A000602.
2. All 621 computed values $a(0), \ldots, a(620)$ agree with the OEIS b-file `b000602.txt` (terms $n = 0..1000$, contributed independently by P. von Brömssen), downloaded and compared term-by-term.
3. As an end-to-end validation of the same code path on a published high-precision target: the singularity-analysis pipeline built on the computed $t(n)$ reproduces Kotesovec's 30-digit constant for A000598, $t(n) \sim C_R \, \alpha^n n^{-3/2}$ with $C_R = 0.5178759064588935369 9\ldots$, to 22 significant digits.
4. The Newton-computed radius of convergence ρ (Proposition 8.3) agrees with OEIS entry A261340 (decimal expansion of ρ, 97 digits, V. Kotesovec 2015) digit-for-digit through all 42 digits computed here.

All matrices in §§4–5 are populated from these certified integers reduced modulo the primes

$$p_1 = 67108859 = 2^{26} - 5, \qquad p_2 = 67108837, \tag{4}$$

chosen below 2^{26} so that all modular row operations stay far inside the exact range of 64-bit integer arithmetic.

3 The certificate method

All non-existence theorems below reduce to the statement that an explicit integer matrix has *full column rank*. The logic is collected here once.

Lemma 3.1 (Modular rank certificate)**.** *Let $M \in \mathbb{Z}^{R\times S}$ and let p be prime. If the reduction $\overline{M} \in \mathbb{F}_p^{R\times S}$ has* $\operatorname{rank}_{\mathbb{F}_p} \overline{M} = S$*, then* $\operatorname{rank}_{\mathbb{Q}} M = S$*; consequently $Mv = 0$ has no nonzero solution $v \in \mathbb{C}^S$.*

Proof. Reduction mod p cannot increase rank: a nonvanishing $S \times S$ minor of $\overline{M}$ lifts to a minor of M not divisible by p, hence nonzero. So $\operatorname{rank}_{\mathbb{Q}} M = S$ and $\ker_{\mathbb{Q}} M = 0$. Since M has rational entries, $\dim_{\mathbb{C}} \ker_{\mathbb{C}} M = \dim_{\mathbb{Q}} \ker_{\mathbb{Q}} M = 0$. □

Remark 3.2. The certificate is one-sided in exactly the safe direction: a *rank deficiency* mod p could be an accident of the prime, but *full rank* mod a single prime is a rigorous proof of non-existence over $\mathbb{C}$. The second prime p_2 in (4) guards only against implementation error, not against a gap in the mathematics. Note also that Lemma 3.1 yields non-existence over $\mathbb{C}$, not merely over $\mathbb{Q}$: closed forms with irrational or transcendental constants (π, $\zeta(3)$, algebraic irrationals, ...) are covered automatically.

Lemma 3.3 (Window reduction: eventual relations imply windowed relations)**.** *Suppose* $\sum_{j=0}^{r} p_j(n)a(n+j) = q(n)$ *holds for all* $n \geq n_0$*, with* $\deg p_j, \deg q \leq d$ *and* $(p_j) \neq 0$*. Put* $s(n) = \prod_{i=1}^{n_0-1}(n-i)$*. Then* $\sum_{j=0}^{r}\big(s\,p_j\big)(n)\,a(n+j) = \big(s\,q\big)(n)$ *holds for* all $n \geq 1$*, with degrees* $\leq d+n_0-1$*. Hence ruling out relations valid from* $n=1$ *in a degree-budgeted region also rules out eventually-valid relations with correspondingly smaller degree.*

Proof. For $n \geq n_0$ multiply the hypothesis by $s(n)$; for $1 \leq n < n_0$ both sides vanish because $s(n) = 0$. □

Lemma 3.4 (Finite windows suffice)**.** *Fix* (r, d) *and let* $S = (r+1)(d+1) + (d+1)$ *count the unknown coefficients of* $(p_0, \ldots, p_r, q)$*. Build* $M \in \mathbb{Z}^{R \times S}$ *whose row indexed by* n $(1 \leq n \leq R)$ *expresses* $\sum_j p_j(n)a(n+j) - q(n) = 0$ *in those unknowns. If* $\operatorname{rank} M = S$ *for some finite* R*, then* $\mathcal{R}(r,d)$ *is false: no relation holds for all* $n \geq 1$*, since it would already have to hold on the window.*

Proof. A relation valid for all $n \geq 1$ is in particular valid for $1 \leq n \leq R$, i.e. its coefficient vector lies in $\ker M = 0$. □

In all runs the window length was $R = \min(620 - r,\ S + 25)$, which always exceeded S by at least 8 rows; ranks were computed by exact Gaussian elimination over $\mathbb{F}_{p_1}$ (and re-verified over $\mathbb{F}_{p_2}$ for three representative cells).

Decoy validation. Before each search the identical code was required to *find* a known relation: the Catalan numbers ($(n+2)C_{n+1} = (4n+2)C_n$; the guesser reports rank deficiency exactly 1 at $(r,d) = (1,1)$), the Catalan generating function ($y = 1 + xy^2$; found at $\deg_y = 2$), and the Thue–Morse sequence (whose 2-kernel closes with 2 states). A search rig that finds the planted relations and reports full rank on $a(n)$ is making a meaningful assertion.

4 No polynomial recurrence: the holonomic perimeter

4.1 Why this class matters: the Wilf–Zeilberger reduction

Definition 4.1 (Hypergeometric-sum closed forms)**.** Let $\mathcal{C}$ be the class of expressions $F(n)$ built, with fixed structure independent of n, from: (i) rational functions of n with constants in $\mathbb{C}$ (the constants may involve π, $\zeta(s)$ at fixed s, Γ at fixed arguments, etc.); (ii) finitely many factors α^n ($\alpha \in \mathbb{C}$ fixed) and $\Gamma(\lambda n + \mu)^{\pm 1}$, binomials $\binom{\lambda n+\mu}{\nu n+\xi}$, with $\lambda, \nu \in \mathbb{Q}$, $\mu, \xi \in \mathbb{C}$; (iii) nested finite sums $\sum_{k=0}^{\lambda n+\mu}$, of fixed depth, of *proper hypergeometric terms* [20, 13] in n and the summation indices, including evaluations ${}_pF_q(\ldots; z_0)$ at fixed z_0 with parameters affine in n; (iv) interleavings by residue classes of n modulo a fixed integer.

Class $\mathcal{C}$ contains the overwhelming majority of expressions matching the operation list of Question 1.1: every binomial-sum formula, every Γ-quotient formula, every terminating-hypergeometric formula in the literature of exact enumeration is in $\mathcal{C}$. (What escapes $\mathcal{C}$: constructs with n inside a *non-affine* or non-arithmetic position, e.g. $\lfloor n\theta \rfloor$ with θ irrational, n^n inside a summand bound, or n-dependent analytic parameters; see §9.)

Theorem 4.2 (Wilf–Zeilberger; Lipshitz closure)**.** *Every* $F \in \mathcal{C}$ *is P-recursive over* $\mathbb{C}$*: there exist* $p_0, \ldots, p_r \in \mathbb{C}[n]$*, not all zero, and* $q \in \mathbb{C}[n]$ *(the inhomogeneous part absorbing degenerate boundary contributions) with* $\sum_{j=0}^{r} p_j(n)F(n+j) = q(n)$ *for all sufficiently large* n*.*

Proof references. Proper hypergeometric multisums are holonomic [20, 13]; P-recursiveness is preserved under sums, products, Cauchy products, and interleaving [19, 9]; Γ-factors with rational λ are handled by passing to arithmetic subprogressions and interleaving. □

Thus a single family of certificates — non-existence of P-recurrences — simultaneously eliminates the entire hypergeometric core of the function class of Question 1.1.

4.2 The certificates

For each pair (r, d), the matrix of Lemma 3.4 (including the inhomogeneous column block for q) was assembled from the certified values $a(1), \dots, a(620)$ and its rank computed over $\mathbb{F}_{p_1}$.

r	d	S	rank	verdict	r	d	S	rank	verdict
2	150	604	604	none	26	18	532	532	none
3	120	605	605	none	31	15	528	528	none
4	100	606	606	none	38	12	520	520	none
5	85	602	602	none	46	10	528	528	none
6	72	584	584	none	56	8	522	522	none
8	55	560	560	none	70	6	504	504	none
10	46	564	564	none	84	5	516	516	none
12	38	546	546	none	96	4	490	490	none
15	31	544	544	none	110	3	448	448	none
18	26	540	540	none	145	2	441	441	none
22	21	528	528	none	200	1	404	404	none
					304	0	306	306	none

Table 1: Non-existence certificates for inhomogeneous P-recurrences $\sum_{j=0}^{r} p_j(n)a(n+j) = q(n)$, $\deg p_j, \deg q \le d$. S is the number of unknown coefficients; "rank" is the exact rank over $\mathbb{F}_{p_1}$ of the window system. Full rank (rank $= S$) proves non-existence over $\mathbb{C}$ (Lemma 3.1). Cells $(2, 150)$, $(10, 46)$, $(304, 0)$ re-verified over $\mathbb{F}_{p_2}$.

Theorem 4.3. *For every (r, d) dominated by an entry of Table 1 (i.e. $r \le r_i$ and $d \le d_i$ for some row i), there is no identity $\sum_{j=0}^{r} p_j(n)\, a(n+j) = q(n)$ valid for all $n \ge 1$ with $p_j, q \in \mathbb{C}[n]$ of degree $\le d$ and $(p_j) \ne 0$. By Lemma 3.3, there is also no such identity valid for $n \ge n_0$ whenever $d + n_0 - 1$ stays within the same region.*

Corollary 4.4. *No $F \in \mathcal{C}$ whose minimal annihilating operator (in the inhomogeneous sense of Theorem 4.2) has order and degree inside the region of Theorem 4.3 can satisfy $F(n) = a(n)$ for all large n. In particular $a(n)$ is given by no binomial/Γ/hypergeometric finite-sum formula of order ≤ 2 and degree ≤ 150, order ≤ 10 and degree ≤ 46, ..., or constant-coefficient recurrence of order ≤ 304 (the last excluding, e.g., every "exponential polynomial" $\sum_i P_i(n)\beta_i^n$ with ≤ 305 terms).*

Remark 4.5 (Calibration of the region)**.** The certified staircase $(r+1)(d+1) \lesssim 600$ is large relative to many familiar examples: the minimal recurrences of numerous classical closed-form sequences (central binomials, Catalan, Motzkin, Apéry, Domb, families of lattice-path and constant-term sequences) have $(r+1)(d+1) \le 40$. Thus any hypergeometric formula for $a(n)$, if it exists within this framework, would have to lie well outside the tested range, in a regime considerably larger than these common examples.

5 No algebraic equation, no first-order ADE

The second classical reservoir of closed forms is Lagrange inversion: if $A(x)$ were algebraic, $a(n)$ would inherit (by Comtet's theorem [5] and the theory of diagonals) precisely the kind of Γ-quotient asymptotics and frequently a finite-sum coefficient formula (Catalan, Motzkin, Schröder, ...). We certify this away directly at the level of A itself.

For a candidate relation $P(x,y) = \sum_{i \le D_x, j \le D_y} c_{ij} x^i y^j$ with $P(x, A(x)) = 0$, impose the vanishing of the coefficients of $x^0, \ldots, x^{620}$ in $P(x, A(x))$: a linear system in the $S = (D_x + 1)(D_y + 1)$ unknowns c_{ij}, with integer entries computed from the certified series (y^j computed by exact modular convolution). Full rank again proves non-existence over $\mathbb{C}$.

D_y	D_x	S	rank	verdict	D_y	D_x	S	rank	verdict
2	190	573	573	none	12	44	585	585	none
3	145	584	584	none	14	38	585	585	none
4	115	580	580	none	17	32	594	594	none
5	95	576	576	none	20	27	588	588	none
6	82	581	581	none	24	22	575	575	none
8	64	585	585	none	28	19	580	580	none
10	52	583	583	none					

Table 2: Non-existence certificates for algebraic relations $P(x, A(x)) = 0$, $\deg_y \le D_y$, $\deg_x \le D_x$. The decoy ($y = 1 + xy^2$ for Catalan) is found by the same code.

Theorem 5.1. *$A(x)$ satisfies no nonzero $P(x, A) = 0$ with $(\deg_y, \deg_x)$ dominated by an entry of Table 2. Moreover, with $P(x,y,z) = \sum_{i,j,k \le 7} c_{ijk} x^i y^j z^k$ (512 unknowns, window of 620 coefficient equations), the system for $P(x, A(x), A'(x)) = 0$ has full rank 512: A satisfies no first-order algebraic differential equation of degree ≤ 7 in each variable.*

Remark 5.2. Theorem 4.3 already implies (within its region) that A is not D-finite with small annihilator, which subsumes small algebraic equations [5]; Theorem 5.1 is nevertheless logically independent and pushes the y-degree much further than the D-finite staircase reaches. The first-order ADE exclusion targets closed forms of Abel/Lambert type ($y' = R(x,y)$) that are not D-finite.

6 No digit formula: automaticity and congruences

A third mechanism producing exact, directly evaluable formulas is *automaticity*/regularity: if $(a(n) \bmod m)$ were k-automatic, or $a(n)$ itself k-regular, then $a(n)$ (or its residues) would be computable by a fixed finite-state device reading the base-k digits of n — the Stern–Brocot, Rudin–Shapiro, paperfolding paradigm [2]. The functional system (2)–(3) has visible Mahler-type structure (the substitutions $x \mapsto x^2, x^3, x^4$), so this possibility deserves a genuine test rather than dismissal.

Recall the k-kernel of a sequence s: the set of subsequences $n \mapsto s(k^e n + r)$, $0 \le r < k^e$, $e \ge 0$. s is k-automatic iff its k-kernel is finite [2, Thm. 6.6.2].

Theorem 6.1. *Within the certified data range, with subsequences compared on overlaps of length ≥ 20:*

1. *the 2-kernel of $(a(n) \bmod 2)$ contains, through depth 4, $1, 3, 7, 15, 31$ pairwise distinct elements — the theoretical maximum $2^{e+1} - 1$ at every depth $e \le 4$; no two of the 31 subsequences agree;*

2. *the 3-kernel of $(a(n) \bmod 3)$ grows $1, 4, 13, 40$ through depth 3 (again maximal, $(3^{e+1}-1)/2$); likewise the 3-kernel of $(a(n) \bmod 2)$;*

3. *for each $m \in \{2, 3, 4, 5, 6, 7\}$, no eventual period $P \le 200$ exists for $(a(n) \bmod m)$ on $250 \le n \le 620$.*

Consequently any 2-DFAO computing $a(n) \bmod 2$ needs ≥ 31 kernel states, any 3-DFAO for $a(n) \bmod 3$ needs ≥ 40; the same Thue–Morse-validated code confirms maximal-rate kernel growth, the canonical signature of non-automaticity.

Remark 6.2 (Honest scope)**.** Items (1)–(3) are finite certificates: they prove kernel *lower bounds* and rule out small periods; unbounded kernel growth (full non-automaticity) is not finitely certifiable and is here an extrapolation — but one supported by maximal growth at every tested depth, in both bases, for both moduli. Note the contrast with genuinely automatic decoys, whose kernels close at depth 1–2. By Cobham's theorem [4] simultaneous 2- and 3-automaticity would force eventual periodicity, which (3) excludes up to period 200.

This rules out small tested versions of a standard route to exact formulas for the *residues* of $a(n)$, let alone $a(n)$ itself; it is also consistent with the expectation that $a(n)$ is not one of the rare "Mahler-automatic" coincidences in which a $(2,3)$-mixed system degenerates to a digit rule. Such degenerations are strongly constrained by the rigidity theory of Mahler equations (Adamczewski–Bell [1], Schäfke–Singer [18]), and the low-complexity rational/algebraic possibilities are excluded here by Theorem 5.1.

7 The labeled problem has a closed form

The next theorem shows that the chemically meaningful constraint — maximum degree 4 — is *not* what blocks Question 1.1.

Theorem 7.1. *For $n \ge 3$, the number of labeled trees on $\{1, \ldots, n\}$ with all degrees ≤ 4 is*

$$L(n) \;=\; \sum_{d=0}^{\lfloor (n-2)/3 \rfloor} \; \sum_{c=0}^{\lfloor (n-2-3d)/2 \rfloor} \frac{n!}{a!\,b!\,c!\,d!} \cdot \frac{(n-2)!}{2^c\, 6^d}, \qquad \begin{array}{l} b = n-2-2c-3d, \\ a = n-b-c-d, \end{array}$$

a fixed-depth (two-index) finite sum of Γ-quotients, i.e. a member of class $\mathcal{C}$ and of the function class of Question 1.1; moreover $L \in \mathcal{C}$ is P-recursive by Theorem 4.2.

Proof. By the Prüfer correspondence, vertex i appears exactly $\deg(i) - 1$ times in the Prüfer word, so the number of labeled trees with prescribed degrees $d_1, \ldots, d_n$ is the multinomial $\binom{n-2}{d_1-1, \ldots, d_n-1}$ [10]. Group the vertices by degree: a vertices of degree 1, b of degree 2, c of degree 3, d of degree 4. The handshake identity $\sum d_i = 2(n-1)$ together with $a+b+c+d = n$ forces $b + 2c + 3d = n - 2$, leaving the two free indices c, d. Choosing which labels receive which degree contributes $n!/(a!\,b!\,c!\,d!)$; the Prüfer multinomial contributes $(n-2)!/\big(0!^{\,a}1!^{\,b}2!^{\,c}3!^{\,d}\big) = (n-2)!/(2^c 6^d)$. Summing over admissible (c, d) gives the formula. It was verified against exhaustive Prüfer-word enumeration for $n \le 8$: $L(1..8) = 1, 1, 3, 16, 125, 1290, 16590, 255920$. $\square$

Remark 7.2. The same argument gives a closed form for *any* fixed degree bound, and analogous exact formulas exist for labeled forests, labeled unicyclic graphs, etc. The entire difficulty of Question 1.1 therefore resides in passing from $L(n)$ to its unlabeled quotient.

8 Localizing the obstruction

8.1 Burnside over partitions: unbounded index dimension

The unlabeled count is the orbit count of the S_n-action on labeled trees:

$$a(n) \;=\; \frac{1}{n!}\sum_{\sigma\in S_n}\mathrm{fix}(\sigma) \;=\; \sum_{\lambda\vdash n}\frac{\mathrm{fix}(\lambda)}{z_\lambda}, \tag{5}$$

where $\mathrm{fix}(\lambda)$ counts degree-≤ 4 labeled trees invariant under a permutation of cycle type λ and z_λ is the usual centralizer order. Equation (5) *is* an exact formula — but its index set is the set of partitions of n: $p(n) \sim e^{\pi\sqrt{2n/3}}/(4n\sqrt{3})$ terms, indexed by integer vectors of *unbounded length*. A fixed-depth finite sum in the sense of Question 1.1 has an index set of bounded dimension growing polynomially. The combinatorial message of Theorems 4.3–6.1 is that no such collapse is found in the tested holonomic, algebraic, or automatic ranges. (For trees, $\mathrm{fix}(\lambda)$ is supported on special λ and is computed in practice through (2)–(3); the partition sum is a faithful image of the substitutions $x \mapsto x^m$ in (2)–(3).)

A useful precedent for such a collapse is instructive.

Remark 8.1 (The Rademacher precedent)**.** $p(n)$ itself is the orbit-count-like coefficient of a natural-boundary generating function, yet possesses the exact formula

$$p(n) = \frac{1}{\pi\sqrt{2}}\sum_{k=1}^{O(\sqrt{n})} A_k(n)\,\sqrt{k}\,\frac{d}{dn}\left(\frac{\sinh\!\big(\frac{\pi}{k}\sqrt{\frac{2}{3}(n-\frac{1}{24})}\big)}{\sqrt{n-\frac{1}{24}}}\right) +\; \Theta, \qquad |\Theta| < \tfrac{1}{2},$$

directly evaluable and exact after rounding [15]. Every step of its proof consumes the $\mathrm{SL}_2(\mathbb{Z})$-modularity of $x^{1/24}/\eta(\tau)$: the circle method needs an exact transformation law at *every* rational point of the boundary. Thus a "natural boundary" per se is not necessarily an obstruction to Question 1.1; the apparent absence of a transformation group is the more relevant difficulty.

8.2 The singular structure of the alkane system

8.2.1 The Otter cancellation and the exponent $-5/2$

Lemma 8.2 (Cycle-index derivative identity)**.** $\dfrac{\partial Z_{S_4}}{\partial t_1} = Z_{S_3}.$

Proof. $\partial_{t_1}\frac{1}{24}(t_1^4 + 6t_1^2t_2 + 3t_2^2 + 8t_1t_3 + 6t_4) \;=\; \frac{1}{24}(4t_1^3 + 12t_1t_2 + 8t_3) \;=\; \frac{1}{6}(t_1^3 + 3t_1t_2 + 2t_3).$ (Equivalently: differentiating the cycle index of S_4 in the fixed-points slot deletes a fixed point, leaving S_3.) □

Proposition 8.3 (Square-root singularity and its cancellation)**.** *T has radius of convergence $\rho =$ 0.3551817423143773928822444736476 . . ., with $a := T(\rho) =$ 2.1174207009536310225162585905 . . . determined by the smooth implicit system*

$$a = 1 + \rho\, Z_{S_3}\big(a, T(\rho^2), T(\rho^3)\big), \qquad \tfrac{\rho}{2}\big(a^2 + T(\rho^2)\big) = 1,$$

and a local expansion $T(x) = a - b\sqrt{1-x/\rho} + c\,(1-x/\rho) + d\,(1-x/\rho)^{3/2} + \cdots$ *with* $b > 0$. *Writing* (3) *as* $A(x) = G\big(x, T(x)\big)$ *with* $G(x,t) = 1 + x\,Z_{S_4}(t, T(x^2), T(x^3), T(x^4)) - \frac{1}{2}\big[(t-1)^2 - (T(x^2)-1)\big]$ *(the inner functions* $T(x^2), T(x^3), T(x^4)$ *being analytic at* $x = \rho$ *since* $\rho^2 < \rho$*), one has the exact cancellation*

$$\frac{\partial G}{\partial t}\Big|_{x=\rho,\, t=a} = \rho\, Z_{S_3}\big(a, T(\rho^2), T(\rho^3)\big) - (a-1) \;=\; 0$$

by Lemma 8.2 and the defining equation of T. *Hence the* $\sqrt{1-x/\rho}$ *term of* T *is annihilated in* A*; the singular expansion of* A *begins at the* $(1-x/\rho)^{3/2}$ *term, and singularity analysis [7, VI–VII] yields, for every* m, *the expansion* (1) *with exponent* $n^{-5/2}$.

Comments on the proof. The square-root type of T at ρ is the standard smooth-implicit-function schema [7, Thm. VII.3] applied to (2), the characteristic condition being the second displayed equation (verified numerically to 55 digits by a two-dimensional Newton iteration in (ρ, a); the system is analytic and nondegenerate there, $\det J \neq 0$). The displayed cancellation is exact and is the celebrated mechanism of Otter [12]: unrooting shifts the polynomial growth from $n^{-3/2}$ (rooted) to $n^{-5/2}$ (unrooted). This is the structural reason the constants of (1) exist; it is also the reason the dominant singularity alone does not reveal a closed-form mechanism — the local behavior is generic square-root, indistinguishable from that of any other simple variety of trees, with all arithmetic specificity pushed into the constants $\rho, a, b, C, c_1, \ldots$, none of which is known to be expressible in classical constants. □

8.2.2 The cascade and the (conjectural) natural boundary

Proposition 8.4 (Singularity cascade; sketch)**.** *The functional equation* (2) *links* $T(x)$ *to* $T(x^2)$ *and* $T(x^3)$. *Beyond* ρ, *the analytic continuation of* T *supplied by* (2) *acquires singularities at points where the* inner *evaluations become singular:* $x^2 = \rho$ *(*$x = \pm\rho^{1/2}$, $|x| \approx 0.5959$*),* $x^3 = \rho$ *(three points,* $|x| \approx 0.7084$*), and inductively at solutions of* $x^{2^a 3^b} = \rho$, *of moduli* $\rho^{1/(2^a 3^b)} \to 1^-$, *with arguments equidistributing on the circle. Granting that no unexpected cancellation kills any of these (each finite instance is numerically checkable; the cancellation at* ρ *itself, Proposition 8.3, is the structurally forced one), the singularities of* T, *hence of* A, *accumulate densely on* $|x| = 1$*: the unit circle is a natural boundary.*

Corollary 8.5 (Conditional global non-holonomicity)**.** *If Proposition 8.4 holds, then* A *has infinitely many singularities; since a D-finite function has only finitely many (the roots of its leading coefficient) [19],* $a(n)$ *is not P-recursive for* any *order and degree, and Corollary 4.4 holds with the region replaced by "everywhere". The certificates of §4 may be viewed as an unconditional finite shadow of this conditional statement.*

8.2.3 No evident transformation theory

The substitution monoid acting in (2)–(3) is generated by $x \mapsto x^2$ and $x \mapsto x^3$: a *mixed Mahler* structure. For pure k-Mahler equations there is a developed Galois/rigidity theory; its flagship results — e.g. a series satisfying both a p-Mahler and a q-Mahler equation with multiplicatively independent p, q must be rational [1, 18] — suggest that a $\{2,3\}$-mixed situation should not be expected to carry a common structure except in special cases. This contrasts with the modular situation of Remark 8.1: η's boundary singularities are all $\mathrm{SL}_2(\mathbb{Z})$-equivalent, while the cascade points $\rho^{1/(2^a 3^b)}\zeta$ of Proposition 8.4 are not known to be organized by a comparable group action. A Rademacher-type exact formula for $a(n)$ would require an exact local expansion at every cascade point *plus* a summable global organization of those expansions; we know of no mathematical technology supplying either ingredient in this setting.

9 Constraints on possible solutions of (A)

Theorem 9.1 (Profile of a hypothetical closed form)**.** *Any F solving Question 1.1 (even only for all $n \geq n_0$, with $n_0 \leq 25$, say) would have to satisfy the following constraints.*

1. ***Outside the WZ core.*** *If $F \in \mathcal{C}$ (Definition 4.1), its minimal (inhomogeneous) annihilating operator must have (r, d) outside the entire staircase of Table 1 (after the degree shift $d \mapsto d + n_0 - 1$ of Lemma 3.3); in particular order ≥ 3 even at degree 150, order ≥ 305 at constant coefficients. No formula of comparable complexity has ever occurred in enumerative combinatorics; conditionally on Proposition 8.4, $F \notin \mathcal{C}$ outright.*

2. ***Not algebraic, not Abelian.*** *The generating function of F can satisfy no $P(x, y) = 0$ within Table 2 and no first-order ADE of degree ≤ 7 per variable.*

3. ***Not a digit formula.*** *$F \bmod 2$ requires ≥ 31 states of 2-kernel, $F \bmod 3$ requires ≥ 40 of 3-kernel; $F \bmod m$ is aperiodic (period > 200) for $m \leq 7$. So F cannot factor through any small finite-state functional of the digits of n.*

4. ***Would need a Burnside collapse.*** *F would have to reproduce (5), a partition-indexed sum of unbounded dimension, within a bounded-depth expression — the only known precedent for which (Rademacher) uses a modular transformation group; no comparable group is currently known for the alkane system or for its pure-Mahler shadow (§8).*

5. ***Exponential precision.*** *By Proposition 1.2, F would have to beat the truncated asymptotic scale (1) by a factor $\alpha^{-n} n^{5/2}$: it would have to encode the constants $\rho, C, c_1, c_2, \ldots$ to unbounded accuracy, i.e. contain the full asymptotic information of the singular expansion, not merely any fixed finite part of it.*

Remark 9.2 (What remains genuinely open)**.** Nothing above excludes: (i) exotic closed forms outside $\mathcal{C}$ (e.g. involving $\lfloor \cdot \rfloor$ of irrational multiples, or special-function evaluations with n in analytically nontrivial positions) — though items (3) and (5) constrain these meaningfully; (ii) a future "mixed-Mahler circle method" supplying a currently absent transformation theory; (iii) larger certified perimeters (the method of §3 scales to thousands of terms with FFT-based convolution and block elimination). One possible deeper route to Question 1.1 would be an exact asymptotic theory for mixed Mahler systems; even a first nontrivial instance of such a theory would be substantial.

A Reproducibility

All computations were performed in Python (exact integer arithmetic; modular linear algebra over $\mathbb{F}_{p_1}, \mathbb{F}_{p_2}$ with 64-bit safe word size; mpmath at 60 decimal digits for the analytic constants only, which feed no certificate). Scripts (in the `alkane_asym` project directory):

`asym.py` Computes $t(0..620)$, $a(0..620)$ from (2)–(3) with exactness assertions; Newton iteration for (ρ, a) to 55 digits; Richardson–Neville extrapolation for C, c_1, c_2, c_3 on two disjoint node sets (agreement: C to 1.7×10^{-18}, c_1 to 10^{-11}, c_2 to 2×10^{-11}).

`solveA.py` The certificate battery of §§4–6 and the labeled-tree verification of §7, including all decoys (Catalan recurrence and algebraic equation; Thue–Morse kernel) and the second-prime re-checks.

`verify_bfile.py` Term-by-term comparison of $a(0..620)$ against the OEIS `b000602.txt` (1001 independent terms): *all* 621 *values agree.*

`crosscheck.py` Reproduction of Kotesovec's published constant for A000598 to 22 digits by the same pipeline.

Indicative cost: series to $n = 620$ in $O(N^2)$ word-multiplications of big integers (seconds); each rank certificate is one dense elimination of a $\leq 650 \times \leq 610$ matrix over $\mathbb{F}_p$ (milliseconds); the full battery runs in well under a minute on a laptop.

References

[1] B. Adamczewski and J. P. Bell, *A problem about Mahler functions*, Ann. Sc. Norm. Super. Pisa Cl. Sci. **17** (2017), 1301–1355.

[2] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.

[3] A. Cayley, *Über die analytischen Figuren, welche in der Mathematik Bäume genannt werden und ihre Anwendung auf die Theorie chemischer Verbindungen*, Ber. Dtsch. Chem. Ges. **8** (1875), 1056–1059.

[4] A. Cobham, *On the base-dependence of sets of numbers recognizable by finite automata*, Math. Systems Theory **3** (1969), 186–192.

[5] L. Comtet, *Calcul pratique des coefficients de Taylor d'une fonction algébrique*, Enseign. Math. **10** (1964), 267–270.

[6] S. R. Finch, *Mathematical Constants*, Cambridge University Press, 2003, §5.6.1 (Otter's tree enumeration constants / chemical isomers).

[7] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009 (alkanes: p. 477–478; transfer theorems: Ch. VI–VII).

[8] F. Harary and R. Z. Norman, *Dissimilarity characteristic theorems for graphs*, Proc. Amer. Math. Soc. **11** (1960), 332–334.

[9] L. Lipshitz, *D-finite power series*, J. Algebra **122** (1989), 353–373.

[10] J. W. Moon, *Counting Labelled Trees*, Canadian Mathematical Monographs, 1970.

[11] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequences A000602, A000598; b-file `b000602.txt` (P. von Brömssen, terms 0..1000). https://oeis.org

[12] R. Otter, *The number of trees*, Ann. of Math. **49** (1948), 583–599.

[13] M. Petkovšek, H. S. Wilf and D. Zeilberger, *A=B*, A K Peters, 1996.

[14] G. Pólya, *Kombinatorische Anzahlbestimmungen für Gruppen, Graphen und chemische Verbindungen*, Acta Math. **68** (1937), 145–254.

[15] H. Rademacher, *On the partition function* $p(n)$, Proc. London Math. Soc. **43** (1937), 241–254.

[16] E. M. Rains and N. J. A. Sloane, *On Cayley's enumeration of alkanes (or 4-valent trees)*, J. Integer Seq. **2** (1999), Article 99.1.1.

[17] R. W. Robinson, F. Harary and A. T. Balaban, *The numbers of chiral and achiral alkanes and monosubstituted alkanes*, Tetrahedron **32** (1976), 355–361.

[18] R. Schäfke and M. F. Singer, *Consistent systems of linear differential and difference equations*, J. Eur. Math. Soc. **21** (2019), 2751–2792.

[19] R. P. Stanley, *Differentiably finite power series*, European J. Combin. **1** (1980), 175–188.

[20] H. S. Wilf and D. Zeilberger, *An algorithmic proof theory for hypergeometric (ordinary and "q") multisum/integral identities*, Invent. Math. **108** (1992), 575–633.